

$$\lambda = \frac{60 \times 10^{-9}}{3}$$

- 17** When both polarisers are aligned, the Amplitude of light from each slit is A.

$$\text{Resultant amplitude} = A + A = 2A$$

$$I_0 \propto (2A)^2 = 4A^2$$

When one of the polarisers are being rotated by  $60^\circ$ ,  
transmitted amplitude from slit  
=  $A \cos 60 = 0.5A$

$$\text{Resultant amplitude} = A + 0.5A = 1.5A$$

$$\text{New } I \propto (1.5A)^2 = 2.25A^2 = \frac{9}{16} I_0$$